

Выполним эквивалентное преобразование схемы и определим комплексные параметры цепи

$$\omega = 2\pi f = 2\pi \cdot 500 = 3141.5927 \text{ rad/s}$$

$$E_1 = \frac{E_{m1}}{\sqrt{2}} e^{j\psi_1} = 21.213 e^{j45^\circ} \text{ В}$$

$$J_2 = \frac{J_{m2}}{\sqrt{2}} e^{j\psi_2} = 14.142 e^{j-45^\circ} \text{ А}$$

$$E_2 = \frac{J_2}{G_{m2}} = \frac{10-j10}{0.2} = 50-j50 \text{ В}, R_{1n2} = \frac{1}{G_{m2}} = \frac{1}{0.2} = 5 \Omega$$

$$E_{eq} = E_1 - E_2 = 15 + j15 - (50 - j50) = -35 + j65 \text{ В}$$

$$Z_{eq,m} = (R_{1n1} + R_{1n2}) + jX_{L1n1} = (5 + 5) + j3 = 10 + j3 \Omega = 10.44 e^{j16.7^\circ} \text{ deg}$$

Рассчитаем эквивалентные сопротивления ветвей приемника

$$X_{L1} = \omega L_1 = 3141.5927 \cdot 0.0024 = 7.54 \Omega, X_{C1} = \frac{1}{\omega C_1} = \frac{1}{3141.5927 \cdot 0.000035} = 9.095 \Omega$$

$$X_{L2} = \omega L_2 = 3141.5927 \cdot 0.0015 = 4.712 \Omega, X_{C2} = \frac{1}{\omega C_2} = \frac{1}{3141.5927 \cdot 0.000045} = 7.074 \Omega$$

$$X_{L3} = \omega L_3 = 3141.5927 \cdot 0 = 0 \Omega, X_{C3} = \frac{1}{\omega C_3} = \frac{1}{3141.5927 \cdot 0.000038} = 8.377 \Omega$$

$$Z_1 = R_1 + jX_{L1} - jX_{C1} = 18 + j7.54 - j9.095 = 18 - j1.55$$

$$Z_2 = R_2 + jX_{L2} - jX_{C2} = 22 + j4.712 - j7.074 = 22 - j2.36$$

$$Z_3 = R_3 + jX_{L3} - jX_{C3} = 27 + j0 - j8.377 = 27 - j8.38$$

Находим эквивалентное сопротивление нагрузки и входное сопротивление цепи

$$Z_{23} = \frac{Z_2 Z_3}{Z_2 + Z_3} = \frac{(22 - j2.36)(27 - j8.38)}{49 - j10.74} = 12.24 - j2.38$$

$$Z_{eq,load} = Z_1 + Z_{23} = 18 - j1.55 + (12.24 - j2.38) = 30.24 - j3.93$$

$$Z_m = Z_{eq,m} + Z_{eq,load} = 10 + j3 + (30.24 - j3.93) = 40.24 - j0.93$$

Определим значения токов

$$I_1 = \frac{E_{eq}}{Z_m} = \frac{-35 + j65}{40.24 - j0.93} = -0.91 + j1.59 \text{ А}$$

$$I_2 = I_1 \frac{Z_3}{Z_2 + Z_3} = -0.91 + j1.59 \cdot \frac{27 - j8.38}{49 - j10.74} = -0.43 + j0.94 \text{ А}$$

$$I_3 = I_1 - I_2 = -0.91 + j1.59 - (-0.43 + j0.94) = -0.47 + j0.66 \text{ А}$$

Выполняем проверку расчетов по законам Кирхгофа

$$\Delta I = 0 + j0, \Delta U_1 = 0 + j0, \Delta U_2 = 0 + j0$$

$$\varepsilon_{KL} = 0\%, \varepsilon_{KIL} = 0\%$$